

BROADBAND MULTILAYER ACTIVE-SLOT POWER DIVIDER

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ABSTRACT: A novel multilayer broadband active-slot power divider is proposed for the first time. As compared to the conventional active multilayer power divider [7], whose 3-dB bandwidth of small-signal gain is typically about 40%, the overall 3-dB bandwidth of the small-signal gain of the proposed multilayer active-slot power divider is about 52%. Through the modification of the T-junction, the proposed power divider is able to provide the characteristics of balance, low loss, and broad band. The proposed device can also provide the possibility of uneven power outputs under appropriate load modifications and amplifications. © 2004 Wiley Periodicals, Inc. *Microwave Opt Technol Lett* 41: 10–12, 2004; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.20029

Key words: multilayer circuit; active power divider; slot coupling

1. INTRODUCTION

The technique of using ground aperture to couple energy has been widely used in the designs of microstrip antennas [1–4]. With this technique, there is no direct connection between the antenna and the feedline; as a result, there is less interference between the antenna and the feedline. In addition to the antenna applications, the ground aperture technique has also been used for vertical transition in multilayer structures [5–6]. Through the slot or aperture, the signal is coupled from a microstrip line in one layer to another microstrip line in the other layer. In order to achieve compact and low-interference design, Hui et al. [7] applied the ground-slot coupling technique to construct a multilayer broadband active-slot power divider. In this design, the adopted T-junction, which splits the input power, is only well matched at the input. As a result, this limits the overall performance of the active power divider. In this paper, a simple but effective modification to alleviate this dependence is proposed. A resistor with a unique value is added across the T-junction's outputs. The modified

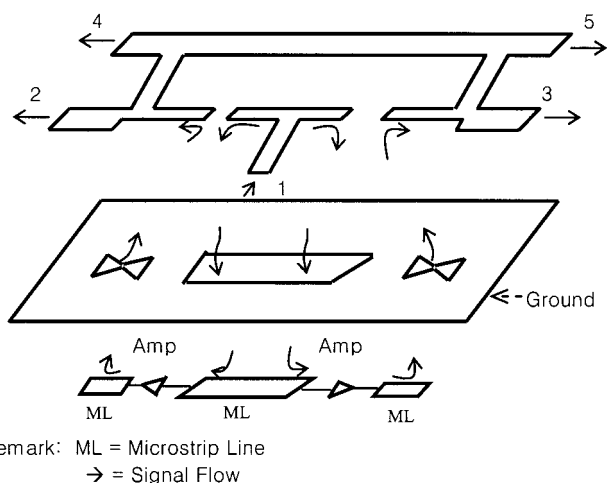


Figure 1 Overall topology of the original multilayer active hybrid-ring

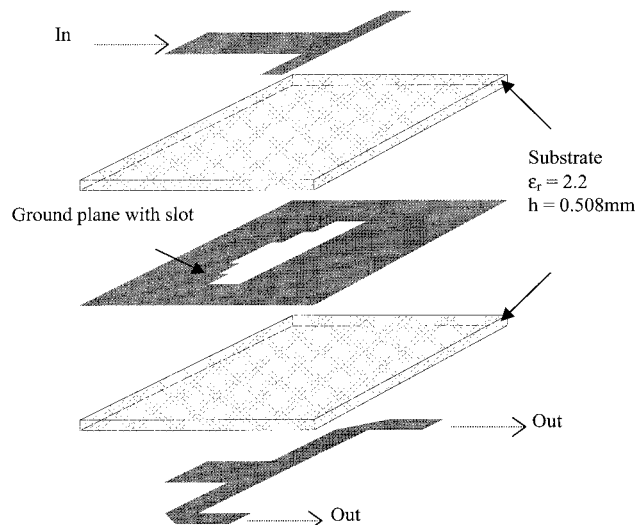


Figure 2 Blow-up of the conventional multilayer active power divider

junction is noted to possess broadband, balanced and low-loss properties.

2. DESIGN OF MULTILAYER ACTIVE SLOT POWER DIVIDER

Before we address the proposed multilayer active-power divider, the basic operation principles of the conventional multilayer active-power divider [7], as shown in Figure 1, must first be reviewed. The T-junction at port 1 acts as a power divider, which splits the energy equally and couples the signal through the slot from the top layer to the bottom layer where the amplifiers are connected. A blow-up view of the T-junction is illustrated in Figure 2. Each split signal is then boosted by the power amplifier module, followed by the wave coupling back through the bowtie slot before leaving at the output ports of the in-phase power divider at the top layer. All the ports are designed with respect to 50Ω load conditions. Because of the load termination at the end of the lower amplifiers' circuit, as expected, matching performs well only at the input end and, therefore, the overall performance of the active power divider will degrade accordingly.

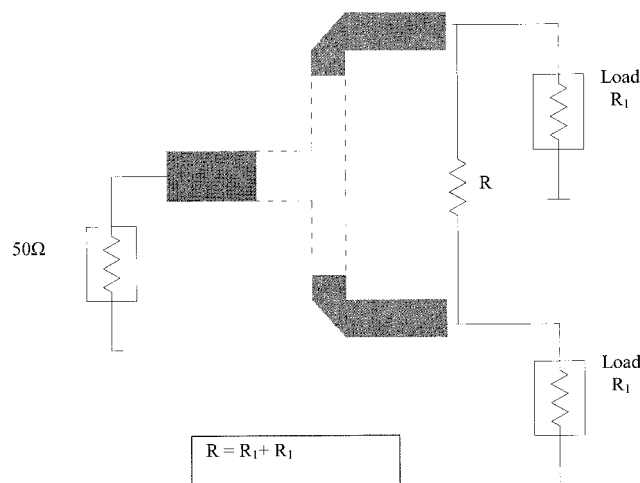


Figure 3 Topology of the lower circuit of the proposed active power divider

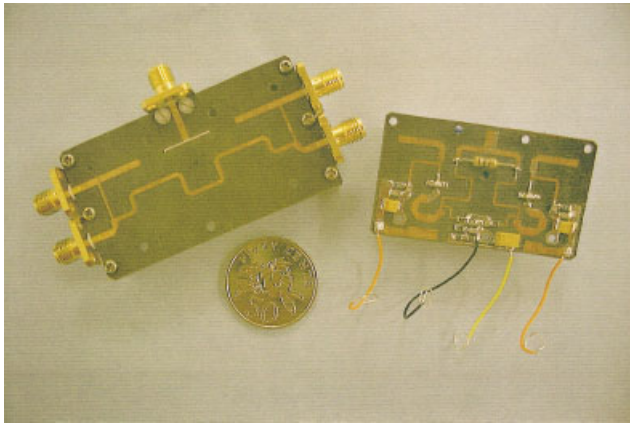
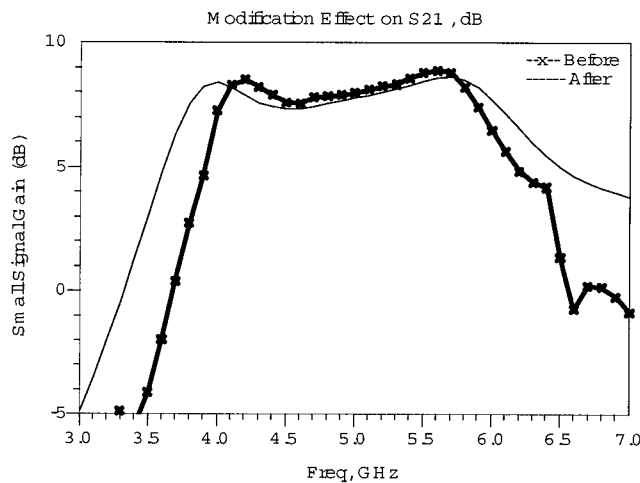


Figure 4 Photo of the proposed multilayer active power divider with top and bottom PCBs. [Color figure can be viewed in the online issue, which is available at www.interscience.wiley.com.]

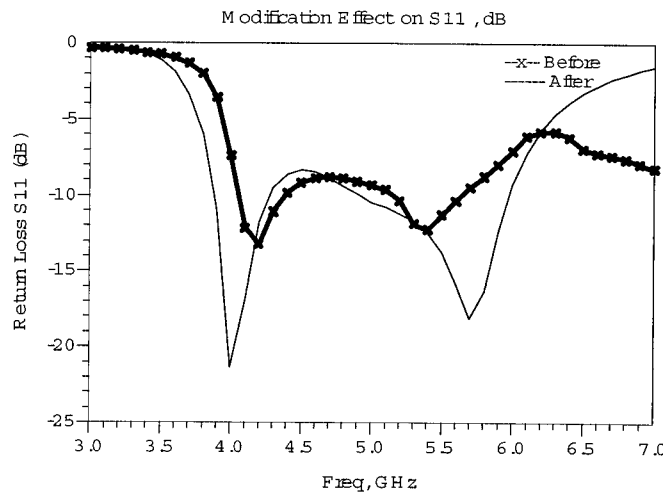
To improve the performance of the coupler, a simple but effective modification is implemented at the output ports of the lower circuit. A resistor, as shown in Figure 3 and the photograph of Figure 4, is next added between the output ports of the lower circuit. Similar to the “Wilkinson power divider” design, based on odd- and even-mode analyses, the value of the resistance is computed to be 100Ω . With the presence of this resistor, all unwanted odd modes will be absorbed by the resistor. The major advantages of this circuit are low-loss, balanced, and broadband characteristics.

3. EXPERIMENTAL RESULTS AND DISCUSSIONS

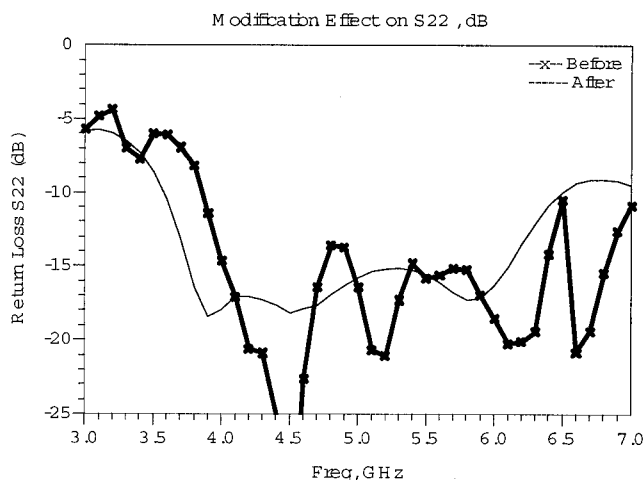
Both the conventional and proposed multilayer active-slot power divider are designed and simulated in IE3D, a commercial software from Zeland Software, Inc. The dimensions of the microstrip lines, as well as the shape and size of the ground slot, are tuned and optimized for the best performance. The two circuits are subsequently fabricated in order to evaluate the performance. Both circuits consist of two printed circuit boards (PCBs) stacking back to back. The top PCB is fabricated with the passive in-phase power divider [8]. The bottom PCB has two single-stage matched ampli-



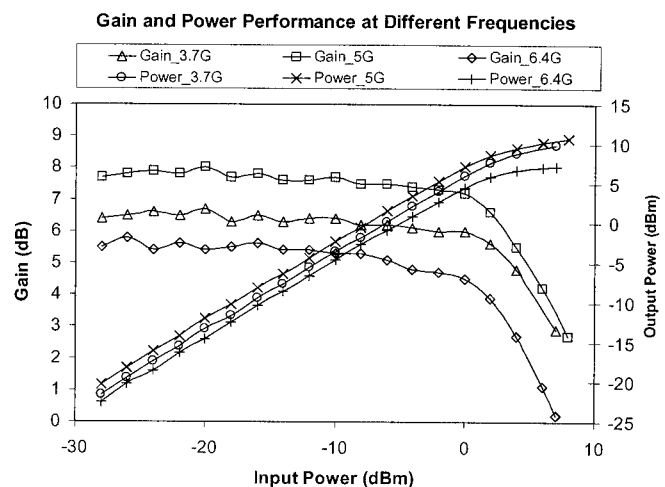
(a)



(b)



(c)



(d)

Figure 5 Comparison of the measurement results before and after improvement: (a) S_{21} gain (dB); (b) S_{11} (dB); (c) S_{22} (dB); (d) gain and power performance

fiers, one for each branch. Both PCBs are made of Duroid substrate with permittivity $\epsilon_r = 2.2$ and thickness 0.508 mm.

Both prototypes are measured by using the commercially available HP8510C Network Analyzer. The output power is obtained by using the Rohde & Schwarz SMP02 Signal Generator and the HP8593E Spectrum Analyzer. Figure 5 compares the measured results of the two fabricated prototypes. The symbol "cross," which is given as "before" in the figure's legend, denotes the performance of the conventional multilayer active-slot power divider, whereas the item "after" in the legend denotes the measured results of the proposed multilayer active-slot power divider. The measurement results of port 3 are very similar to that of port 2; therefore, for the sake of brevity, they are not shown here.

As observed from the figures, the proposed active-slot power divider exhibits superior performance, compared to that of the conventional multilayer power divider. As noted from the S_{21} plot, a 52% bandwidth for the 3-dB bandwidth small-signal gain has been achieved with the proposed multilayer active power divider whereas the conventional multilayer power divider only has 40% bandwidth. This improvement in performance is mainly due to the fact that the added resistor helps to absorb the reflected power from the load termination of the lower circuit. Besides the good gain performance, good return loss of less than -15 dB for the S_{22} has also been obtained.

The power performance at 3.7, 5, and 6.4 GHz are plotted in Figure 5(d). Comparing to results in [7], there is not much difference in the P_{1dB} compression point. Also noted from the power plot in Fig. 4(d), the number of amplifiers used, as well as the PCB's substrate materials, actually limits the output power and the gain. Moreover, in our experimentation, it is noticed that the bandwidth of the amplifier unit governs the overall output bandwidth.

4. CONCLUSION

A novel multilayer active power divider has been designed, simulated and fabricated. Compared to the conventional, multilayer active power divider in [7], it is noted that a broader 3-dB bandwidth for the small-signal gain of about 52% has been obtained. The proposed modification also improves the output return loss of the overall active slot power divider.

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A BROADBAND CPW-FED INDUCTIVE SLOT ANTENNA

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ABSTRACT: In this paper, a broadband coplanar waveguide (CPW)-fed inductive slot antenna is presented. It consists of a conventional CPW inductive slot antenna and a CPW inductively coupled slot antenna in the feeding network. The experimental results provide the characteristic of excellent enhancement of impedance bandwidth ($VSWR < 2$). The impedance bandwidth of this broadband antenna can reach about 4.5 GHz (2.35–6.85 GHz). Details of the antenna design and experimental results are discussed. © 2004 Wiley Periodicals, Inc. *Microwave Opt Technol Lett* 41: 12–14, 2004; Published online in Wiley InterScience (www.interscience.wiley.com). DOI 10.1002/mop.20030

Key words: CPW-fed slot antenna; broadband antenna; slot antenna; inductive

1. INTRODUCTION

Coplanar waveguide (CPW)-fed slot antennas have been widely used for various applications because they are compatible with monolithic microwave integrated circuits (MMICs) and active devices [1]. In addition, the bandwidth of CPW-fed slot antennas is larger than conventional microstrip antenna [2, 3]. They have enough bandwidth to support modern communication systems. In [2], a PBG structure is incorporated into the feeding network of the CPW-fed loop slot antenna, which not only allows harmonic suppression, but also impedance bandwidth enhancement. Conventional CPW-fed slot antennas take many kinds of forms, such as the capacitive loop slot antenna [2], capacitive slot antenna [3], capacitive folded slot antenna [4], and inductive slot antenna. A new antenna exhibits good performance by incorporating two different forms CPW-fed slot antennas [5]. This new antenna consists of an inductive slot antenna and a capacitive slot antenna. Two antennas generate two resonant frequencies such that the first resonant frequency is adjacent to the second resonant frequency

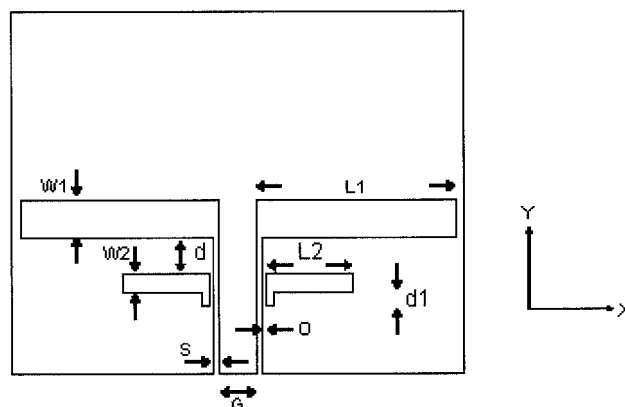


Figure 1 Geometry of the broadband CPW-fed inductive slot antenna